

Diseases caused by Fungi and Protists *

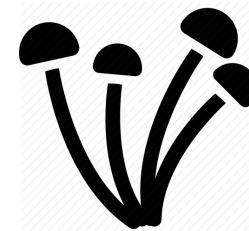
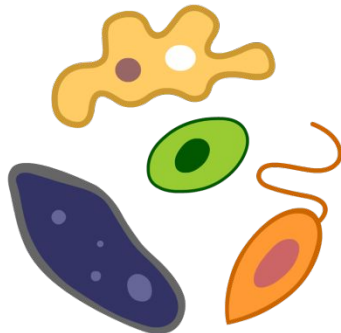
Key Words: Rose black spot, athletes foot, malaria, mosquito, pathogen, vector

DO NOW: Unscramble these sentences.

Sentence 1: need – reproduce – viruses – cells – to

Sentence 2 : larger – viruses – are – bacteria – than - and – toxins - release

Sentence 3: moist – fungus – conditions – grow – in – need – to - order

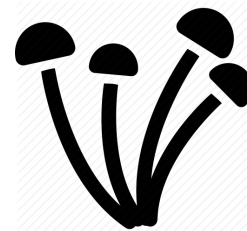
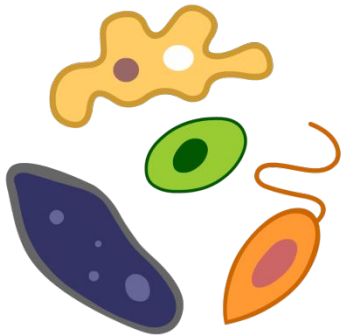


Review

Sentence 1: Viruses need cells to reproduce

Sentence 2 : Bacteria are larger than viruses and release toxins

Sentence 3: Fungus need moist conditions in order to grow



Diseases caused by Fungi and Protists *

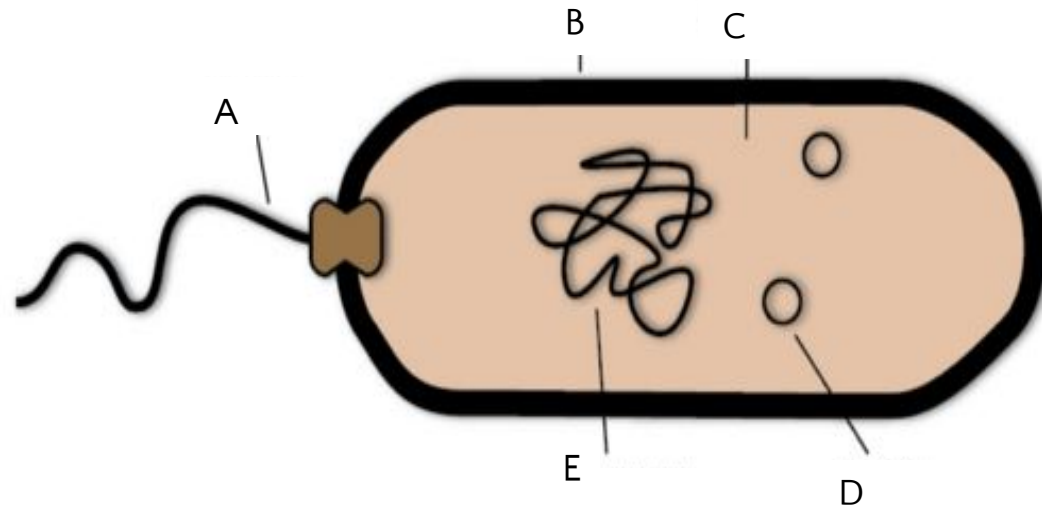
Key Words: Rose black spot, athletes foot, malaria, mosquito, pathogen, vector

DO NOW: Answer these three review questions below.

1. Name part B and D
[2 marks]

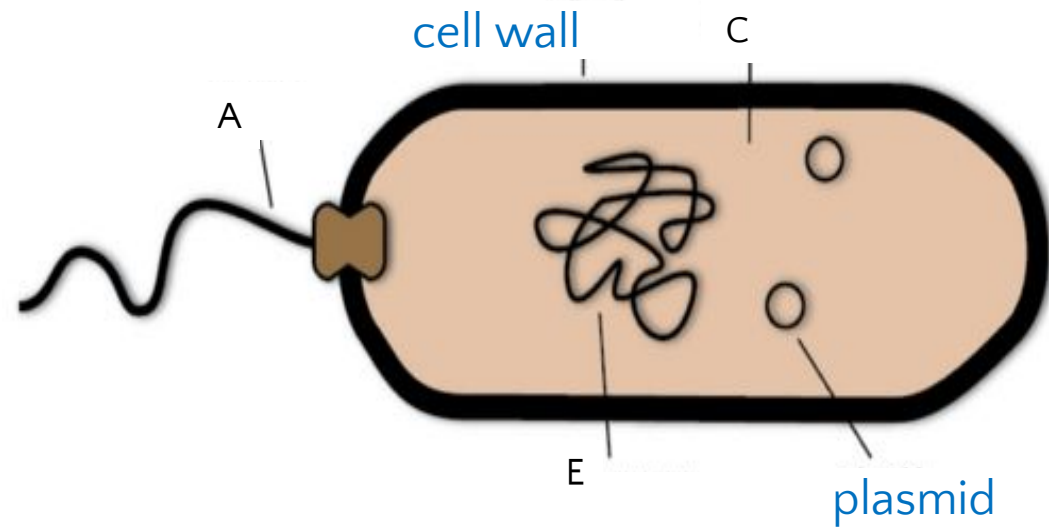
2. Compare the features of a bacterium and a virus [3]

3. State how gonorrhoea and salmonella spreads and how to prevent and treat it. [4]



Review

1. Name part B and D
[2 marks]



2. Compare the features of a bacterium and a virus [3]

Bacteria are bigger than viruses [1]

Viruses reproduce inside of cells, bacteria reproduce by binary fission [1]

Bacteria are cells that can move independently
viruses can't move by themselves [1]

3. State how gonorrhoea and salmonella spreads and how to prevent or treat it. [4]

Spread: Gonorrhoea = sexual contact [1]

Salmonella = contaminated food, poor hygiene [1]

Prevention: Gonorrhoea = condoms, antibiotics [1]

salmonella = good hygiene, vaccinate chickens [1]

Bacterial Diseases

*

Key Words: Salmonella, Gonorrhoea, sexually-transmitted-disease, symptom, antibiotic, food

Starter: Let's do a quiz on what we learnt last lesson. 1 □ 10

TIME FOR A

QUIZ!



Quiz:

1. Which bacterial disease causes yellow discharge from the reproductive organs? a) HIV b) gonorrhoea c) Salmonella d) TMV
2. Which disease is also called food poisoning?
3. What drug can be used to treat bacteria diseases?
4. _____ don't work on viruses because viruses reproduce.
5. Name a symptom of salmonella.
6. Which disease leads to discolouration of plant leaves?
7. Which diseases is passed by sexual contact or exchange of fluids?
8. Which animal is vaccinated so we don't get salmonella?
9. Bacteria releases _____ so that makes us feel ill.
10. Which disease has the symptom red skin rash and fever?

LOb: Understand how pathogenic fungi and protists cause disease.

Keywords:

Rose black spot, athlete's foot, malaria, mosquito, pathogen, vector

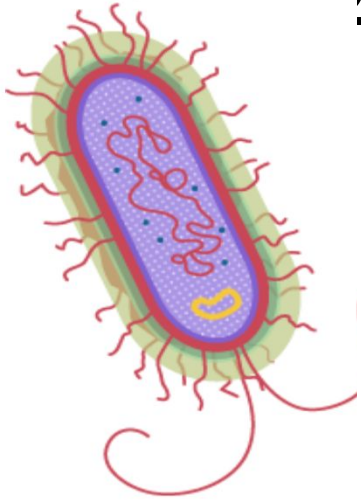


Success Criteria:

- Describe fungi and explain rose black spot
- Describe a protist and explain malaria

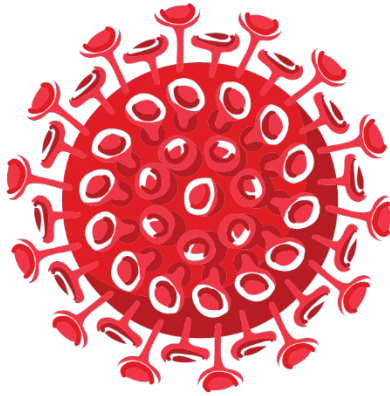
Did you know... 198 million people had malaria in 2013. 584,000 were fatal.

Bacteria, Viruses, Fungi and Protists



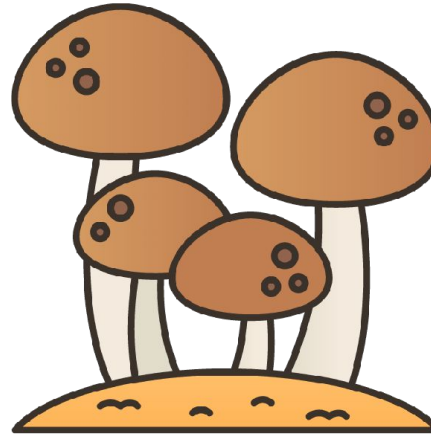
BACTERIA

Reproduce rapidly
inside the body.
Release toxins
that make us ill



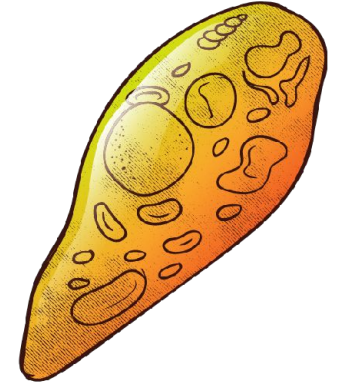
VIRUS

Live reproduce
inside human cells
damaging cells.
Rapidly reproduce



FUNGI

Multicellular
organisms that
prefer damp/moist
environments



PROTIST

Microorganisms that
have features
belonging to plants
animals and fungi

Pathogens: Are microorganism that causes
infectious disease.

Make notes on the features of fungi.

WATCH AND

LEARN



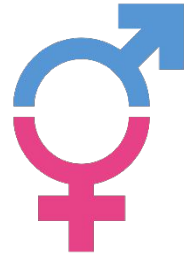
Review

Not animals
or plants

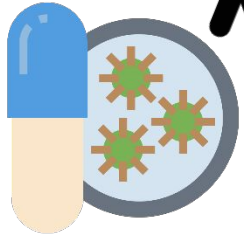
- ☐ multicellular
- ☐ unicellular



They team up with
plants to help each
other grow



Reproduce sexually
or asexually



Source of
antibiotics

Fungi

Used to make food
and drink



They don't make
their own food
from photosynthesis



**They like
moist environments**

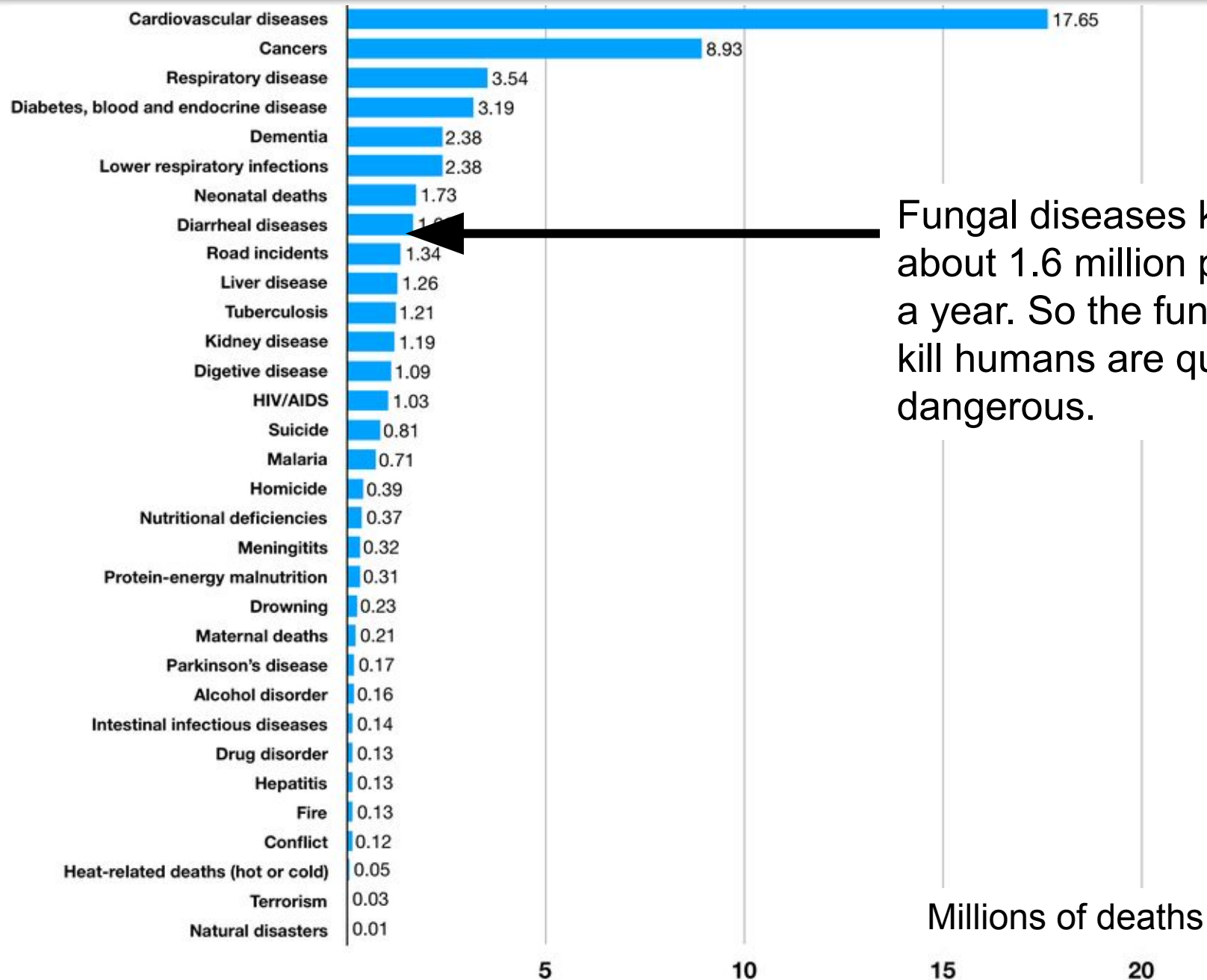


Fungi

A group of organisms preferring moist/ damp conditions that feed on organic matter and produce spores for example mould, yeast, mushrooms and toad stalls.



There are relatively few fungal diseases that kill humans.



Fungal diseases kills about 1.6 million people a year. So the fungi that do kill humans are quite dangerous.

In **plants**, however, **fungus diseases are common** and can be devastating. Huge areas of crops are lost to fungal diseases every year. Bananas are at real risk!!

WATCH AND

LEARN

AFTER ALL KNOWLEDGE

POWER!



What do you think has happened to this leaf?

Rose Black Spot



Activity: Complete your worksheet table as you watch the video.

WATCH AND

LEARN

AFTERALL KNOWLEDGE IS

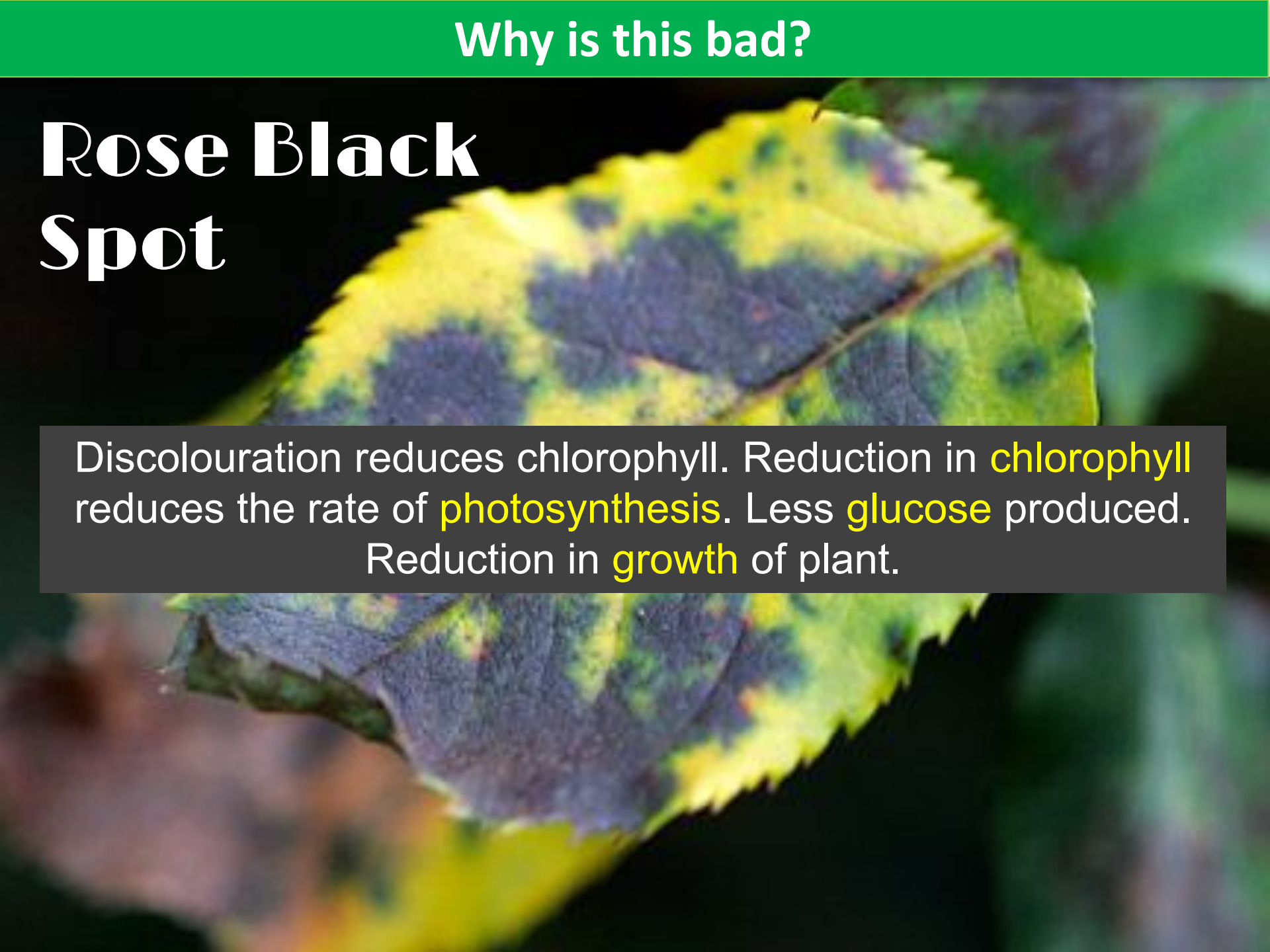
POWER!



Why is this bad?

Rose Black Spot

Discolouration reduces chlorophyll. Reduction in **chlorophyll** reduces the rate of **photosynthesis**. Less **glucose** produced.
Reduction in **growth** of plant.



LOb: Understand how pathogenic fungi and protists cause disease.

Keywords:

Rose black spot, athlete's foot, malaria, mosquito, pathogen, vector



Success Criteria:

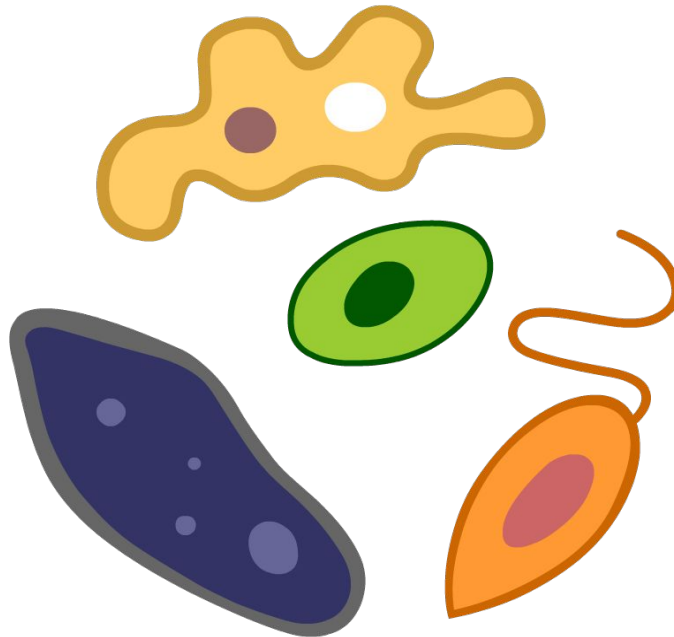
1. Describe fungi and explain rose black spot

2. Describe a protist and explain malaria

Did you know... 198 million people had malaria in 2013. 584,000 were fatal.

Protists

A type of **single celled organism** that causes disease; normally these diseases require a **vector** to **transfer** the protist **to the host**.



WHAT ARE THEY?

Unicellular ✓

Reproduce
independently ✓

Communicable ✓

Eukaryotic ✓

Producers ✓

Large ✓

Moist
environment ✓

Can move ✓

Microscopic ✓

Consumers ✓

30

Activity: Use the AQA biology book p84-85 to complete the table fill for malaria.



Activity: Complete your worksheet table as you watch the video.

WATCH AND

LEARN

AFTERALL KNOWLEDGE IS

POWER!



**Review: In the envelope you will have some statements.
some are about malaria some are not. Sort the
statements.**



Review

true

Caused by plasmodium protist living in mosquitos

Symptoms include fever and if untreated cause death

Transmitted by a vector. A vector is an organism that spreads a communicable disease

We prevent the disease spreading using nets and insecticide

Young and elderly people in poor countries, travellers, pregnant women and those with HIV/AIDS are most at risk

Treatment includes anti-malaria drugs and blood transfusion

There is no vaccination for malaria



Activity: Write a leaflet alerting people to the disease of malaria. Make sure you use the correct information.

Success criteria

- > Comment on symptoms, prevention
- Cause, treatment
- > Pictures
- > Catchy title



LOb: Understand how pathogenic fungi and protists cause disease.

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Learning Outcomes:

✓ Describe fungi and explain rose black spot

✓ Describe a protist and explain malaria

✓ Practice exam questions

Did you know... 198 million people had malaria in 2013. 584,000 were fatal.

keyfacts

💡 **Fungi** like **moist** conditions and spread by **direct contact**.
Protists spread using a **vector** (mosquitos).

💡 **Rose black spot:** symptoms= **black spots** on leaves. How it spreads: **direct contact**. Prevention/treatment: remove infected leaves. Fungicide.



💡 **Malaria:** symptoms= fever. How it spreads: **vector (mosquitos)**. Prevention/treatment: Remove standing water, use **mosquito nets** to control spread.

Plenary

We are going to create a memory chain.
Each person adds to the chain and remembers
And vocalises what the previous person has said.

Fungi are multicellular and...

Malaria is caused by plasmodium and...



Plenary

A stylized purple notepad with a black outline. It features a header box at the top, a large central drawing area, and four horizontal lines at the bottom for writing.

One key word that is stuck in your mind from today

A picture to describe today's learning

A sentence to describe your learning today